

Trigonometry

1. Trigonometry

1.1. Domain/Range of trig functions

Identities	Domain	Range
$\sin \theta$	$\mathbb{R}$	$[-1, 1]$
$\cos \theta$	$\mathbb{R}$	$[-1, 1]$
$\tan \theta$	$\mathbb{R} - \left\{(2n + 1)\frac{\pi}{2}\right\}$	$\mathbb{R}$
$\cot \theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R}$
$\csc \theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R} - (-1, 1)$
$\sec \theta$	$\mathbb{R} - \left\{(2n + 1)\frac{\pi}{2}\right\}$	$\mathbb{R} - (-1, 1)$

1.2. Trigonometric Formulas

1. $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
2. $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
3. $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
4. $\cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$
5. $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$
6. $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$
7. $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$
8. $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$
9. $\sin(A + B) \cdot \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

$$10. \cos(A + B) \cdot \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$11. \sin C + \sin D = 2 \sin\left(\frac{C + D}{2}\right) \cdot \cos\left(\frac{C - D}{2}\right)$$

$$12. \sin C - \sin D = 2 \cos\left(\frac{C + D}{2}\right) \cdot \sin\left(\frac{C - D}{2}\right)$$

$$13. \cos C + \cos D = 2 \cos\left(\frac{C + D}{2}\right) \cdot \cos\left(\frac{C - D}{2}\right)$$

$$14. \cos C - \cos D = 2 \sin\left(\frac{C + D}{2}\right) \cdot \sin\left(\frac{D - C}{2}\right)$$

$$15. \sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$16. \cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$17. \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$18. \cot 2A = \frac{\cot^2 A - 1}{2 \cot A}$$

$$19. \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$20. \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$21. \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$22. \cot 3A = \frac{\cot^3 A - 3 \cot A}{3 \cot^2 A - 1}$$

$$23. \tan^2 A = \frac{1 - \cos 2A}{1 + \cos 2A}$$

1.3. Value of θ

$$1. \sin \theta = \sin \alpha \implies \theta = n\pi + (-1)^n \alpha$$

$$2. \cos \theta = \cos \alpha \implies \theta = 2n\pi \pm \alpha$$

$$3. \tan \theta = \tan \alpha \implies \theta = n\pi + \alpha$$

$$4. \sin \theta = 0 \implies \theta = n\pi$$

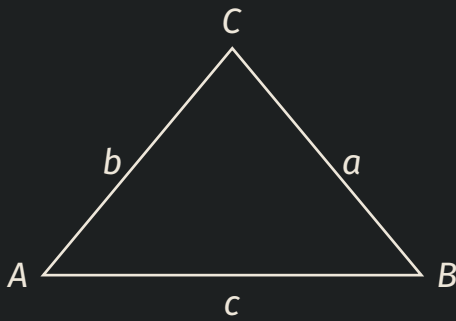
$$5. \cos \theta = 0 \implies \theta = n\pi + \frac{\pi}{2} = (2n + 1)\frac{\pi}{2}$$

$$6. \tan \theta = 0 \implies \theta = n\pi$$

$$7. \sin \theta = 1 \implies \theta = 2n\pi + \frac{\pi}{2} = (4n + 1)\frac{\pi}{2}$$

$$8. \cos \theta = 1 \implies \theta = 2n\pi$$

2. Properties of a triangles



2.1. Area of a triangle

$$1. \Delta = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B$$

$$2. \Delta = \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{a+b+c}{2}$$

2.2. Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{where circumradius, } R = \frac{abc}{4\Delta}$$

2.3. Law of Cosines:

$$1. a^2 = b^2 + c^2 - 2bc \cos A \implies \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$2. b^2 = a^2 + c^2 - 2ac \cos B \implies \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$3. \ c^2 = a^2 + b^2 - 2ab \cos C \implies \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

2.4. Projection Rule:

1. $a = b \cos C + c \cos B$
2. $b = c \cos A + a \cos C$
3. $c = a \cos B + b \cos A$

2.5. Napier's analogy:

1. $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}$
2. $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot \frac{B}{2}$
3. $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot \frac{C}{2}$

2.6. Half Angle Formulas:

1. $\sin\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{bc}}$
2. $\cos\left(\frac{A}{2}\right) = \sqrt{\frac{s(s-a)}{bc}}$
3. $\tan\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{r}{(s-a)}$

where inradius, $r = 4R \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right)$

3. Inverse Trigonometry

3.1. Domain/Range of inverse trig functions

Identities	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$

3.2. Inverse Trigonometric Formulas

- $\sin^{-1} x = \csc^{-1}\left(\frac{1}{x}\right)$
- $\cos^{-1} x = \sec^{-1}\left(\frac{1}{x}\right)$
- $\tan^{-1} x = \cot^{-1}\left(\frac{1}{x}\right), x > 0$
 $= \cot^{-1}\left(\frac{1}{x}\right) - \pi, x < 0$
 $= 0, x = 0$
- $\csc^{-1} x = \sin^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \sec^{-1} x$
- $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \csc^{-1} x$
- $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1}(x)$
- $\sin^{-1}(-x) = -\sin^{-1} x$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$
- $\tan^{-1}(-x) = -\tan^{-1} x$

$$10. \cot^{-1}(-x) = \pi - \cot^{-1} x$$

$$11. \csc^{-1}(-x) = -\csc^{-1} x$$

$$12. \sec^{-1}(-x) = \pi - \sec^{-1} x$$

$$13. \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$14. \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

$$15. \csc^{-1} x + \sec^{-1} x = \frac{\pi}{2}$$

$$16. \sin^{-1} x \pm \sin^{-1} y = \sin^{-1} \left(x\sqrt{1-y^2} \pm y\sqrt{1-x^2} \right)$$

$$17. \cos^{-1} x \pm \cos^{-1} y = \cos^{-1} \left(xy \mp \sqrt{1-x^2}\sqrt{1-y^2} \right)$$

$$18. \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right), xy < 1$$

$$= \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), xy > 1, x > 0, y > 0$$

$$= -\pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), xy > 1, x < 0, y < 0$$

$$19. \tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

$$20. \tan^{-1} x \pm \tan^{-1} y \pm \tan^{-1} z = \tan^{-1} \left(\frac{x \pm y \pm z \mp xyz}{1 \mp xy \mp yz \mp zx} \right)$$

$$21. 2 \sin^{-1} x = \cos^{-1} (1 - 2x^2) = \sin^{-1} (2x\sqrt{1-x^2})$$

$$22. 2 \cos^{-1} x = \cos^{-1} (2x^2 - 1)$$

$$23. 2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$24. 3 \sin^{-1} x = \sin^{-1} (3x - 4x^3)$$

$$25. 3 \cos^{-1} x = \cos^{-1} (4x^3 - 3x)$$